Backend — Factory Game UML (Unity)

«MonoBehaviour» GameManager
-_beltTrack: BeltTrack «SerializeField»
-_itemSpawner: ItemSpawner «SerializeField»
-_grinderPrefab: GameObject «SerializeField»
-_welderPrefab: GameObject «SerializeField»
-_painterPrefab: GameObject «SerializeField»
-_managers: List~IManager~
-_placedMachines: List~Machine~
+OnEvent(e: GameEvent): void
+StartGame(): void
+PauseGame(): void
+ResetGame(): void
+PlaceMachine(type: MachineType, pos: int): void
+LevelUpMachine(pos: int): void
+RemoveMachine(pos: int): void
+LevelUpBelt(): void
«MonoBehaviour» Machine
-_beltTrack: BeltTrack «SerializeField»
-_upgradeInterval: float «SerializeField»
-_level: int
-_cooldown: float
-_upgradeAmount: float
-_type: MachineType
+Level: int «get, private set»
+UpgradeInterval: float «get, private set»
+UpgradeAmount: float «get, private set»
+InstallPrice: float «get»
+LevelUpPriceCoeff: float «get»
+Configure(beltTrack: BeltTrack): void
+LevelUp(): void
+GetMachineType(): MachineType
#SetMachineType(type: MachineType): void
-UpgradeItem(item: Item): void
-OnTriggerStay(other: Collider): void
+GetTargetStat(): StatType
+CanCauseDefect(): bool
«MonoBehaviour» ItemSpawner
-_beltTrack: BeltTrack «SerializeField»
-_itemPrefab: GameObject «SerializeField»
-_baseAP: float «SerializeField»
-_baseDU: float «SerializeField»
-_baseSP: float «SerializeField»
-_spawnInterval: float «SerializeField»
-_timer: float
-_level: int
+BaseAP: float «get, private set»
+BaseDU: float «get, private set»
+BaseSP: float «get, private set»
+SpawnInterval: float «get, private set»
+ShouldSpawn(): bool
+SpawnItem(): void
+LevelUp(): void
+OnBeltTrackLevelChanged(s: IBeltTrackLevelSubject): void
«MonoBehaviour» BeltTrack
-_sellManager: SellManager «SerializeField»
-_trackLength: float «SerializeField»
-_sellBlock: BlockBase «SerializeField»
-_level: int
-_machineSpaces: int
-_speed: Vector3
-_items: List~Item~
-_itemPositions: Dict~Item, float~
-_observers: List~IObserver~
+Level: int «get, private set»
+MachineSpaces: int «get, private set»
+Speed: Vector3 «get, private set»
+TrackLength: float «get, private set»
+LevelUpPrice: float «get»
+AddItem(item: Item): void
+RemoveItem(item: Item): void
+GetNearestItem(xPosition: float): Item
+LevelUp(): void
+GetMachineSpaces(): int
+NotifyBeltTrackLevel(): void
+OnSellBlockReached(item: ISellable): void
«MonoBehaviour» Grinder
+GetTargetStat(): StatType
+CanCauseDefect(): bool
«MonoBehaviour» Painter
+GetTargetStat(): StatType
+CanCauseDefect(): bool
«MonoBehaviour» Welder
+GetTargetStat(): StatType
+CanCauseDefect(): bool
2: Update() GetNearestItem
+ OnTriggerStay() alphaEff
Splendor
AttackPower
Durability
instantiates prefab
upgrades
0..*
_placedMachines
1
_itemSpawner
1
1
_beltTrack
1
1
_beltTrack
1
_managers
_items
1
0..*
Subscribe
_sellManager
1
1
IsGameOver / ResetStatus
«MonoBehaviour» Item
-_attackPower: float
-_durability: float
-_splendor: float
-_isDefective: bool
-_position: Vector3
+AttackPower: float «get, protected set»
+Durability: float «get, protected set»
+Splendor: float «get, protected set»
+IsDefective: bool «get, protected set»
+Position: Vector3 «get, set»
+SetValue(ap: float, du: float, sp: float): void
+MoveItem(dp: Vector3): void
+Upgrade(stat: StatType, amount: float): void
+GetStats(): Dict~StatType, float~
+CalculatePrice(spMult: float): float
+CalculateDefectChance(): float
+MakeDefective(): void
+SellItem(): void
«MonoBehaviour» StandardItem
IUpgradable
+Upgrade(stat: StatType, amount: float): void
+GetStats(): Dict~StatType, float~
ISellable
+IsDefective: bool «get»
+CalculatePrice(spMult: float): float
+SellItem(): void
IManager
+OnEvent(e: GameEvent): void
«MonoBehaviour singleton» SellManager
-_instance: SellManager
-_apMultiplier: float
-_spMultiplier: float
-_activeEvent: EventType
+Instance: SellManager «get»
+ApMultiplier: float «get, set»
+SpMultiplier: float «get, set»
+ActiveEvent: EventType «get, private set»
+OnEvent(e: GameEvent): void
+SellItem(item: ISellable): void
+ApplyFine(item: ISellable, p: float): void
«MonoBehaviour singleton» RoundManager
-instance: RoundManager
-_roundNumber: int
-_apThreshold: float
-_checkInterval: float
-_bonus: float
-_penalty: float
-_soldItemAPs: List~float~
-_observers: List~IObserver~
+Instance: RoundManager «get»
+RoundNumber: int «get, private set»
+ApThreshold: float «get, private set»
+CheckInterval: float «get, private set»
+Bonus: float «get, private set»
+Penalty: float «get, private set»
+AverageAP: float «get»
+OnEvent(e: GameEvent): void
+StartRound(): void
-EndRound(): void
+ScaleNextRound(): void
+NotifyRound(): void
Subscribe / Publish(RoundEndEvent)
ModifyMoney
ModifyMoney / AddBrandPoints
Subscribe / Publish(ItemSoldEvent)
«use» param
0..*
«singleton» EventBus
-_instance: EventBus
-_listeners: List~IGameEventListener~
+GetInstance(): EventBus
+Subscribe(l: IGameEventListener): void
+Unsubscribe(l: IGameEventListener): void
+Publish(e: GameEvent): void
1
_listeners
0..*
dispatches
IGameEventListener
+OnEvent(e: GameEvent): void
GameEvent
+Timestamp: float «get»
RoundEndEvent
+AvgAP: float «get»
+Passed: bool «get»
ItemSoldEvent
+Price: float «get»
+IsDefective: bool «get»
+Splendor: float «get»
+AttackPower: float «get»
MarketEvent
+EventType: EventType «get»
BankruptcyEvent
+BarLevel: float «get»
«MonoBehaviour singleton» FactoryStatus
-_instance: FactoryStatus
-_factoryStatusValue: Dict~FactoryStatusType, float~
-_observers: List~IObserver~
+Money: float «get»
+BrandLevel: float «get»
+BrandPoints: float «get»
+SplendorMultiplier: float «get»
+BankruptcyBar: float «get»
+GetInstance(): FactoryStatus
+Threshold(): float
+ModifyMoney(v: float): void
+AddBrandPoints(sp: float): void
+UpdateBankruptcyBar(delta: float): void
+IsGameOver(): bool
+ResetStatus(): void
+NotifyFactoryStatus(type: FactoryStatusType, arg: UIUpdateArgs): void
IFactoryStatusSubject (FRONTEND.puml).
Update():
-Money/2000*dt , dt/60
IBeltTrackLevelSubject + ISellBlockObserver (FRONTEND.puml).
_sellBlock(FrontEnd BlockBase)
Start() SellManager.Instance — DESIGN.md
Pure C#: EventBus, GameEvent
MonoBehaviour: Item , Machine , BeltTrack,
ItemSpawner, SellManager, RoundManager, GameManager, FactoryStatus
FactoryStatusSubject, IRoundSubject,
IBeltTrackLevelSubject, ISellBlockObserver
FrontEnd/ObserverPattern/ — FRONTEND.puml
StatType
AttackPower
Durability
Splendor
MachineType
Grinder
Welder
Painter
EventType
None
Boom
Recession
FactoryStatusType
Money
BrandLevel
BrandPoints
BankruptcyBar
Operation
Addition
Multiplication
Assignment

FrontEnd — Factory Game UML (Unity)

Assets/Scripts/FrontEnd/ (namespace)
ObserverPattern/ ISubject/IObserver
(FactoryStatus, RoundManager, BeltTrack)
— BACKEND.puml
«MonoBehaviour» BlockSelect
-selectObj: GameObject «SerializeField»
-m_UiPanels: List~UIPanelMapping~ «SerializeField»
-uiDict: Dict~BlockUIType, UIPanelBase~
-currentBlock: BlockBase
-HandleSelection(): void
-SetCurrentUI(uiType: BlockUIType): void
-SelectNewBlock(newBlock: BlockBase): void
-DeselectCurrentBlock(): void
-CloseAllUI(): void
UIType
Text
Slider
Button
raycast
uiDict
1
0..*
BlockUIType
Null
MachineSelect
MachineModify
TrackModify
«MonoBehaviour» CameraMove
-speed: float
-initPos: Vector3
«MonoBehaviour» BlockBase
#Observers: List~IObserver~ «get»
+UIType(): BlockUIType
+SelectBlock(): void
+UnselectBlock(): void
+NotifyBlock(): void
«MonoBehaviour» UIPanelBase
-panel: GameObject «SerializeField»
-_targetBlock: BlockBase
+TargetBlock: BlockBase «get, protected set»
+OpenUI(block: BlockBase): void
+CloseUI(): void
+SetSelectOj(selectObj: GameObject): void
+OnBlockChanged(block: IBlockSubject): void
+OnButtonSelected(button: IUIPanelButtonSubject): void
«MonoBehaviour» FactoryStatusUI
-factoryStatusTypes: List~FactoryStatusType~ «SerializeField»
-uiViews: List~UIView~ «SerializeField»
-uiDict: Dict~FactoryStatusType, UIView~
+OnFactoryStatusChanged(type: FactoryStatusType, arg: UIUpdateArgs): void
«MonoBehaviour» RoundUI
-roundNumTxt: UIView «SerializeField»
-timeLimitTxt: UIView «SerializeField»
-targetApTxt: UIView «SerializeField»
-currentApTxt: UIView «SerializeField»
+OnRoundChanged(s: IRoundSubject, arg: UIUpdateArgs): void
«MonoBehaviour» SellBlock
+UIType(): BlockUIType
+NotifySellBlock(item: ISellable): void
-OnTriggerEnter(other: Collider): void
ObserverPattern
IObserver
+OnNotify(subject: ISubject): void
ISubject
+RegisterObserver(o: IObserver): void
+UnregisterObserver(o: IObserver): void
+NotifyObservers(): void
«MonoBehaviour» BeltBlockManager
-beltBlockPrefab: GameObject «SerializeField»
-beltBlocks: List~BeltBlock~
-beltTrack: BeltTrack
+generateBeltBlock(): void
+OnBeltTrackLevelChanged(s: IBeltTrackLevelSubject): void
«MonoBehaviour» TrackModifyUI
-_trackLevelTxt: UIView «SerializeField»
-_trackLevelUpPriceTxt: UIView «SerializeField»
+OnBlockChanged(block: IBlockSubject): void
+OnButtonSelected(button: IUIPanelButtonSubject): void
«MonoBehaviour» MachineModifyUI
-_machineTypeTxt: UIView «SerializeField»
-_levelTxt: UIView «SerializeField»
-_levelUpPayTxt: UIView «SerializeField»
+OnBlockChanged(block: IBlockSubject): void
+OnButtonSelected(button: IUIPanelButtonSubject): void
«MonoBehaviour» MachineSelectUI
-buttonPrefab: GameObject «SerializeField»
-buttons: List~UIPanelButtonBase~
+OnButtonSelected(button: IUIPanelButtonSubject): void
«MonoBehaviour» UIPanelButtonBase
-_observers: List~IObserver~
+OnButtonClick(): void
1
MachineLevelUp / SellMachine
beltBlocks
0..*
«MonoBehaviour» BeltBlock
-_machine: Machine
+Machine: Machine «get, set»
+MachineName: string «get»
+MachineLevel: float «get»
+MachineLevelUpPrice: float «get»
+MachineSellPrice: float «get»
+UIType(): BlockUIType
+CreateMachine(type: MachineType): bool
+SellMachine(): void
+MachineLevelUp(): bool
-PayMoney(pay: float): bool
CreateMachine
«MonoBehaviour» TrackBlock
-_beltTrack: BeltTrack
+TrackLevel: int «get»
+TrackLevelUpPrice: float «get»
+UIType(): BlockUIType
+TrackLevelUp(): void
+PayMoney(pay: float): bool
TrackLevelUp
GetInstance().RegisterObserver
uiViews
1
0..*
1
Instance.RegisterObserver
4
detects
FactoryStatus (Backend)
«MonoBehaviour» UIView
+SetValue(arg: UIUpdateArgs): void
RoundManager (Backend)
ISellable (Backend)
IFactoryStatusObserver
+OnFactoryStatusChanged(type: FactoryStatusType, arg: UIUpdateArgs): void
IRoundObserver
+OnRoundChanged(s: IRoundSubject, arg: UIUpdateArgs): void
ISellBlockSubject
+NotifySellBlock(item: ISellable): void
IBlockSubject
+NotifyBlock(): void
ISellBlockObserver
+OnSellBlockReached(item: ISellable): void
IUIPanelButtonObserver
+OnButtonSelected(button: IUIPanelButtonSubject): void
IBlockObserver
+OnBlockChanged(block: IBlockSubject): void
IBeltTrackLevelObserver
+OnBeltTrackLevelChanged(s: IBeltTrackLevelSubject): void
IBeltTrackLevelSubject
+NotifyBeltTrackLevel(): void
IRoundSubject
+NotifyRound(): void
IFactoryStatusSubject
+NotifyFactoryStatus(type: FactoryStatusType, arg: UIUpdateArgs): void
IUIPanelButtonSubject
«MonoBehaviour» MachineSelectButton
+MachineType: MachineType «get, set»
+Initialize(type: MachineType, observer: IUIPanelButtonObserver): void
«MonoBehaviour» MachineModifyButton
«MonoBehaviour» TrackModifyButton_LevelUp
LevelUp
GetMachine(type)
Instantiate/LevelUp
ModifyMoney
«MonoBehaviour singleton» MachineManager
-machineInfoList: MachineInfoList «SerializeField»
-machineDict: Dict~MachineType, GameObject~
+Instance: MachineManager «get»
+GetMachine(type: MachineType): GameObject
BeltTrack (Backend)
Machine (Backend)
FactoryStatus (Backend)
«MonoBehaviour» MachineModifyButton_Levelup
«MonoBehaviour» MachineModifyButton_Sell
1
1
«ScriptableObject» MachineInfoList
-machineInfos: List~MachineInfo~ «SerializeField»
+Count: int «get»
+GetMachineAt(i: int): MachineInfo
1
0..*
«ScriptableObject» MachineInfo
-type: MachineType «SerializeField»
-machine: GameObject «SerializeField»
+GetMachineType(): MachineType
+GetMachine(): GameObject
«use» param
«MonoBehaviour» SliderUIView
+SetValue(arg: UIUpdateArgs): void
«MonoBehaviour» ButtonUIView
+SetValue(arg: UIUpdateArgs): void
UIUpdateArgs
+Value: object «get»
wraps (RoundUI)
«MonoBehaviour» TextUIView
+Prefix: string «get, protected set»
+Suffix: string «get, protected set»
+SetValue(arg: UIUpdateArgs): void
BankruptcyUIView
BrandPointUIView
SliderUpdateArgs
+MaxValue: float «get, set»
TextUpdateArgs
RoundParameters
+roundNum: int «get»
+timeLimit: float «get»
+targetAp: float «get»
+currentAp: float «get»
RoundUIView_TargetAP
RoundUIView_CurrentAP
MachineModifyUIView_MachineType
MachineModifyUIView_Level
MachineModifyUIView_LevelUpPrice
TrackModifyUIView_Level
TrackModifyUIView_LevelUpPrice
MoneyUIView
BrandLevelUIView
RoundUIView_RoundNum
RoundUIView_TimeLimit